Width Confounding in Label-Free Cross-Architecture Model Selection: Diagnosis and Subspace Correction

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Abstract

Selecting a pretrained encoder without target labels is a standard preprocessing step in transfer learning. RankMe, a widely used label-free score, was validated primarily within architecture families. We show that its effective-rank score is strongly confounded by penultimate-layer width when heterogeneous architecture families are compared: under an isotropic random-feature model, RankMe effective rank scales as $\Theta(d)$, and empirically its cross-family rankings track width rather than transfer quality. We propose Subspace-Corrected Ranking (SCR), which augments target spectral information with the relative geometry between each candidate's target features and its features on a fixed reference dataset. Its parameter-free member $\mathrm{SSR}^{\ddagger}$ subtracts log subspace alignment from log effective rank. On a common support of 22 pool–task units from MV, VTAB, and EX, $\mathrm{SSR}^{\ddagger}$ reduces mean regret from 0.173 (RankMe) to 0.069; the paired difference is -0.1043 (target-name-clustered 95% CI $[-0.167, -0.040]$, $p = 0.00118$). $\mathrm{SSR}^{\ddagger}$ is intended for cross-family selection, not as a universal replacement for RankMe: RankMe remains preferable within a family. A five-task MV experiment further shows that $\mathrm{SSR}^{\ddagger} \rightarrow$RankMe can retain similar regret while reducing forward evaluations, which we treat as a proof of concept rather than a general cost guarantee.

1 Introduction

Transfer learning starts with a choice: which pretrained encoder to fine-tune on the target task. When candidate encoders share the same architecture but differ in pretraining recipe, the choice can be made cheaply via a label-free score on the encoder's penultimate features. The dominant such score, *RankMe* [8], computes the effective rank $r_{\mathrm{eff}}(H)$ of the feature matrix and ranks candidates by $\log r_{\mathrm{eff}}$. In the within-architecture setting where RankMe was validated, the penultimate-layer dimension d is constant across candidates and the score reflects representational quality.

The realistic deployment setting is different. A practitioner faced with the timm [30] or torchvision model zoo chooses between fundamentally different backbones—ResNet [10], ViT [5], ConvNeXt [16], EfficientNet [26], and others—whose penultimate dimensions vary by an order of magnitude. RankMe was validated on within-family variants [8]; however, practitioners selecting among heterogeneous backbones on model hubs such as timm [30] and Hugging Face Hub naturally apply it pool-wide, without the within-family stratification its validation assumed. In this cross-architecture setting, we show that RankMe-based selection fails systematically. The rankings it produces are dominated by penultimate-layer width: on a pool of 132 pretrained variants spanning 35 architectures, RankMe selects the widest available variant on every task and performs worse than uniform

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random selection. We observe the same RankMe-specific confound in the other heterogeneous pools studied here.

The mechanism is a consequence of RankMe's target-only spectral form under feature isotropy. A corrected Marchenko–Pastur argument shows that RankMe effective rank scales as $\Theta(d)$ at fixed positive aspect ratio, so $\log r_{\text{eff}}$ equals $\log d$ plus an aspect-ratio-dependent constant asymptotically. This formal statement is specific to RankMe; rotation invariance alone is insufficient to conclude that an arbitrary score is monotone in d.

The correction is relational rather than target-spectral. We propose *Subspace-Corrected Ranking* (SCR), which compares each candidate encoder's target features with reference-dataset features extracted by that same encoder. The score is invariant to a common change of representation basis, but unlike RankMe it is not determined by the target singular spectrum alone. Across the candidate pool, SA_k correlates strongly negatively ($\rho_{\mathrm{Sp}} = -0.959$) with the fraction of class-mean variance outside the reference subspace. This is mechanistic evidence, not a formal guarantee of downstream accuracy. The default member, $\mathrm{SSR}^{\ddagger} = [\log r_{\text{eff}} - \log \mathrm{SA}_k]/\log C$, is designed for cross-family selection; RankMe is retained for within-family variant selection.

Contributions. **(i) Diagnosis.** We establish RankMe's $\Theta(d)$ scaling under an isotropic random-feature model and document strong width confounding across heterogeneous model pools. **(ii) Correction.** We propose SCR, a family of label-free relational scores that incorporate within-candidate target–reference subspace geometry, with empirical mechanistic support on 168 architecture–task pairs. **(iii) Scope-aware pipeline.** We use $\mathrm{SSR}^{\ddagger}$ for cross-family selection and RankMe within a family; the resulting five-task MV cost–quality result is a proof of concept. **(iv) Benchmark.** Four reproducible evaluation pools covering 52 task units across supervised and SSL/multimodal pretraining paradigms, with backbone widths spanning $d \in [192, 4096]$. The pools are deliberately calibrated at three distinct $d_{\max}/d_{\min}$ ratios ($5.3\times$, $10.7\times$, $21.3\times$), exposing the regimes in which existing corrections succeed or fail and providing a diagnostic testbed for future label-free transferability scores. Pre-computed features, all result tables, and scoring code are released for community use.

2 Related Work

Transferability scores. RankMe [8] ranks pretrained encoders by the effective rank of penultimate features, using $p_i = \sigma_i / \sum_j \sigma_j$; our focus is the heterogeneous-pool setting beyond its original within-family validation. VNE [13] uses the entropy of normalised squared singular values, but we do not extend our RankMe-specific theorem to VNE. Official classification DiSCO [33] uses target ground-truth labels to construct class-conditional quantities and is therefore reported as a label-dependent comparator, not a zero-label baseline. Other label-requiring scores include LogME [31], H-score [2], NCTI [29], LEEP [19], N-LEEP [15], PACTran [4], OTCE [27], GBC [22], TransRate [12], NCE [28], and Task2Vec [1]. Transfer learning more broadly is surveyed in Pan and Yang [21].

Subspace alignment and random matrix theory. Subspace alignment [7] was introduced for unsupervised domain adaptation via principal-angle geometry on the Grassmann manifold [9]. Neural network representation similarity via CKA and SVCCA [14, 24] uses related geometric ideas to compare layers; we use it differently here, as a label-free measure of generic-vs-specific feature structure relative to a fixed reference dataset encoded by each candidate. The Marchenko–Pastur law [18] has been applied to characterise weight matrices in deep networks [17]; we apply it directly to feature matrices to derive the width-confound mechanism.

Two-stage and budgeted model selection. Staged screening followed by precise evaluation is a standard design in neural architecture search and Bayesian hyper-parameter optimisation. In source-free transferability estimation [29, 31], the same logic supports a cross-architecture stage at low cost followed by a within-family stage; we evaluate this design and a mixed-budget extension as five-task proofs of concept.

3 Setup

Pools. We use four architecturally distinct pools whose descriptions are summarised in Table 1 (full per-pool details in Appendices B–E). **MV pool** (multi-variant pool; 132 timm [30] variants, 35 multi-variant architectures, 5 tasks) is the only pool with multiple variants per architecture and is therefore the only pool on which two-stage pipelines are evaluated. **VTAB pool** [32] uses the 24 architectures with complete cached features and LP values across all 8 tasks; STL-10 is used only as the reference in the common-support evaluation, leaving 7 targets. **SSL pool** (8 SSL/multimodal encoders including CLIP [23], DINOv2 [20], DINO [3], and MoCo [11]; 29 tasks) uses pretraining-objective diversity on a smaller set of backbones. **EX pool** (extended supervised pool; 51 timm variants from 35 canonical architectures including VGG-16, d=4096; 10 tasks) covers the same $d_{\max}/d_{\min}$=21.3× width span as MV pool with broader task coverage. Across the four pools we cover 52 task units and widths $d \in [192, 4096]$.

Table 1: Four architecturally distinct evaluation pools used throughout the paper. The $d_{\max}/d_{\min}$ spread drives the severity of the width confound and the choice of correction (see §7).

Pool	Archs	Variants	Tasks	d range	$d_{\max}/d_{\min}$
MV pool (multi-variant)	35	119 (multi-var.) / 132 (full)	5	[192, 4096]	21.3×
VTAB pool	24	24	8	[192, 2048]	10.7×
SSL/multimodal pool	8	8	29	[384, 2048]	5.3×
EX pool (supv.)	35	51	10	[192, 4096]	21.3×

Scores. Effective rank r_{eff} follows RankMe [8]: $p_i = \sigma_i / \sum_j \sigma_j$ and $r_{\text{eff}} = \exp(-\sum_i p_i \log p_i)$. For each candidate encoder m, SA_k compares its target features with its STL-10 features; both are produced by encoder m and therefore share dimension d_m. We use uncentred feature matrices, default $k = 20$, and clip positive quantities below at 10^{-12} before taking logarithms. Different candidates' subspaces are never compared directly. Official classification DiSCO and the other label-requiring methods are reported separately from the zero-label RankMe–$\text{SSR}^\ddagger$ comparison.

Linear-probe ground truth. We use the held-out accuracy (lp_acc) of ℓ_2-regularised logistic regression with C=1 on a single 80/20 split. Strategy quality is reported as *regret*, the gap to the per-task oracle accuracy. Confidence intervals are obtained by task-level bootstrap with n_{boot}=10 000 resamples; paired bootstrap for head-to-head comparisons.

4 The Width Confound

We isolate the failure mechanism with two formal statements. Both proofs are in Appendix A.

Lemma 1 (Marchenko–Pastur scaling)**.** *Let $H \in \mathbb{R}^{n \times d}$ have i.i.d. rows from $\mathcal{N}(0, \sigma^2 I_d)$. As $n, d \to \infty$ with $\gamma = n/d$ fixed, $r_{\text{eff}}(H)/d \to c(\gamma) > 0$ almost surely, where $c(\gamma)$ depends only on the aspect ratio. In particular $r_{\text{eff}}(H) = \Theta(d)$.*

Proposition 2 ($\text{SSR}^\ddagger$ is relational and jointly basis-invariant)**.** *For candidate encoder m and $Q \in O(d_m)$, jointly replacing $(H_{m,T}, H_{m,R})$ by $(H_{m,T}Q, H_{m,R}Q)$ leaves their principal angles, SA_k, and $\text{SSR}^\ddagger$ unchanged. Nevertheless, $\text{SSR}^\ddagger$ is not determined by the singular values of $H_{m,T}$ alone: it additionally depends on the relative geometry of the target and reference right-singular subspaces.*

Lemma 1 is a RankMe-specific statement; rotation invariance alone does not imply monotonicity in d for an arbitrary score. Proposition 2 explains how $\text{SSR}^\ddagger$ adds information absent from the target spectrum while remaining independent of an arbitrary common basis choice. Empirically, on EX the mean Kendall correlation between RankMe and d is $\tau = +0.617$ across 10 tasks, confirming strong width concordance.

Empirical confound across pools. RankMe remains strongly aligned with width: mean per-task $\rho_{\text{Sp}}(\text{RankMe}, d)$ is 0.61 on MV, 0.879 on the complete 24-architecture VTAB pool, and 0.815 on SSL. Width itself is not a reliable quality proxy: its mean correlation with LP accuracy is -0.10, $+0.23$, and -0.45, respectively. On MV, RankMe selects `vgg16` (d=4096) on all five tasks and has regret 0.125 worse than random. On VTAB, its mean regret (0.110) is also slightly worse than

random (0.106), despite the positive average width–accuracy correlation. Thus width can dominate RankMe without reliably identifying the best transferable representation.

Table 2: Width confound across MV, VTAB, and SSL pools (EX pool diagnosis data in Appendix E). RankMe correlates strongly with d (large positive ρ_{Sp}) yet d itself does not predict quality (near-zero or negative $\rho_{\mathrm{Sp}}(d, \mathrm{lp_acc})$), confirming that d is a confound rather than a signal.

Pool	Arch., tasks	$\rho_{\mathrm{Sp}}(\mathrm{RankMe}, d)$	$\rho_{\mathrm{Sp}}(d, \mathrm{lp_acc})$	RankMe regret	Random regret
MV pool	35, 5	$+0.61$	-0.10	$+0.319$	$+0.194$
VTAB	24, 7	$+0.879$	$+0.23$	$+0.110$	$+0.106$
SSL	8, 29	$+0.815$	-0.45	$+0.103$	$+0.066$

VTAB per-task breakdown. Table 3 uses the same 24 candidates for every non-reference target. The RankMe–width correlation ranges from 0.838 to 0.915, while regret varies substantially by task.

Table 3: VTAB per-task width confound on the complete 24-architecture support. STL-10 is excluded because it is the reference dataset.

Task	$\rho_{\mathrm{Sp}}(\mathrm{RankMe}, d)$	Regret	RankMe pick
CIFAR-10	$+0.865$	$+0.050$	efficientnet_b3 (d=1536)
CIFAR-100	$+0.869$	$+0.235$	efficientnet_b3 (d=1536)
DTD	$+0.877$	$+0.095$	resnet101 (d=2048)
Fashion-MNIST	$+0.886$	$+0.045$	wide_resnet50_2 (d=2048)
Flowers-102	$+0.838$	$+0.165$	efficientnet_b3 (d=1536)
Pets	$+0.902$	$+0.070$	mobilenetv3_large (d=1280)
SVHN	$+0.915$	$+0.110$	efficientnet_b3 (d=1536)
Mean	$\mathbf{+0.879}$	$\mathbf{+0.110}$	—

Severity is driven by inter-architecture width variance. The MV pool failure is sharpest because the pool spans $d \in [192, 4096]$, with VGG-16 occupying the high-d tail. Removing the three VGG variants ($\sim 3\%$ of the pool) shifts Flat-RankMe regret by -0.175 LP, bringing it close to random (Table 6); the same removal does not change $\mathrm{SSR}^{\ddagger}$'s observed regret. At a per-task level (Table 4), Flat-RankMe selects `vgg16` on every MV pool task while $\mathrm{SSR}^{\ddagger}$ selects a candidate within 0.07 LP of the oracle on $4/5$ tasks.

Table 4: Per-task breakdown of Flat-RankMe and Flat-$\mathrm{SSR}^{\ddagger}$ on MV pool's multi-variant subset. RankMe selects `vgg16` on every task; $\mathrm{SSR}^{\ddagger}$ recovers task-appropriate architectures.

Task	Oracle	Random	Flat-RM	Flat-$\mathrm{SSR}^{\ddagger}$	RM picks	$\mathrm{SSR}^{\ddagger}$ picks	Oracle arch
CIFAR-10	0.970	0.785	0.550	0.870	vgg16	resnet50	convnext_base
CIFAR-100	0.825	0.484	0.180	0.670	vgg16	resnet152	convnext_base
DTD	0.740	0.564	0.445	0.720	vgg16	convnext_base	convnext_base
EuroSAT	0.955	0.881	0.765	0.905	vgg16	efficientnet_b3	densenet201
SVHN	0.615	0.430	0.570	0.450	vgg16	efficientnet_b3	densenet169

An independent confirmation via spectral truncation. Recomputing r_{eff} on the top-D truncated singular spectrum of each candidate provides an independent check that the confound lives in the singular-value tail. At D=256 on MV pool, the per-task Spearman correlation $\rho_{\mathrm{Sp}}(r_{\mathrm{eff},D=256}, d)$ drops from $+0.63$ (full spectrum) to $+0.06$, and Flat-RankMe regret drops from $+0.319$ to $+0.094$, comparable to $\mathrm{SSR}^{\ddagger}$. Truncation is not a viable correction for two-stage use, however: small-d architectures (ViT-tiny, DeiT-tiny, both d=192) are not truncated at all and retain the width ordering; the corresponding stage-1 regret on the multi-variant subset is $+0.204$, more than twice that of $\mathrm{SSR}^{\ddagger}$ (Appendix E).

Related empirical failures. On the MV pool, LogME and the trace form of H-score also select `vgg16` on every task and degrade when the VGG variants are present (Table 5), whereas LEEP and NCTI do not show the same pattern. We report this only as an empirical observation: our formal

width-confounding result is specific to RankMe under the stated isotropic model and does not cover arbitrary spectral or rotation-invariant scores.

Table 5: MV multi-variant-support flat comparison (119v×5t). Columns report per-task mean regret with and without the three VGG variants in the pool. Feature-spectrum scores (RankMe, LogME, H-score) all select `vgg16` when present and degrade similarly; class-distribution scores (LEEP, NCTI) and the proposed $\mathrm{SSR}^{\ddagger}$ are robust.

Method	With VGG	Without VGG	Δ	Type
Random	+0.194	+0.193	−0.001	—
Flat-RankMe	+0.319	+0.144	−0.175	label-free
Flat-LogME	+0.319	+0.550	+0.231	label-based
Flat-H-score	+0.319	+0.470	+0.151	label-based
Flat-NCTI	+0.162	+0.155	−0.007	label-based
Flat-$\mathrm{SSR}^{\ddagger}$	**+0.098**	**+0.098**	**0.000**	**label-free**
Flat-LEEP	+0.038	+0.038	0.000	label-based

Table 6: Sensitivity of flat strategies to the presence of VGG-class features (d=4096) in MV pool. Columns report mean LP across the 5 tasks. RankMe and the trace-spectrum label-based scores are dominated by VGG when present; $\mathrm{SSR}^{\ddagger}$ and class-distribution scores (LEEP, NCTI) are robust.

Method	With VGG	Without VGG	Δ	Type
Random	0.629	0.630	+0.001	—
Flat-RankMe	0.502	0.677	+0.175	label-free
Flat-NormRankMe ($-\log d$)	0.600	0.449	−0.151	label-free
Flat-$\mathrm{SSR}^{\ddagger}$	**0.723**	**0.723**	**+0.000**	label-free
Flat-LogME	0.502	0.271	−0.231	label-based
Flat-H-score	0.502	0.351	−0.151	label-based
Flat-LEEP	0.783	0.783	+0.000	label-based
Flat-NCTI	0.659	0.666	+0.007	label-based

5 Subspace-Corrected Ranking

Geometric premise. For candidate encoder m, let $V_{m,\mathrm{ref},1:k}$ be the top-k right singular vectors of its features $H_{m,D_{\mathrm{ref}}}$ on the fixed reference dataset, and write $P = V_{m,\mathrm{ref},1:k} V_{m,\mathrm{ref},1:k}^{\top}$, $Q = I - P$. Define

$$f_{\mathrm{spe}}(H, H_{\mathrm{ref}}) := \frac{\mathrm{tr}(Q\Sigma_B Q)}{\mathrm{tr}(\Sigma_B)},$$

the fraction of between-class covariance lying outside the reference subspace. f_{spe} measures how much of a candidate's discriminative variance is architecture-specific rather than shared with a generic encoder. On the complete VTAB support (24×7 = 168 architecture–task pairs), the subspace alignment $\mathrm{SA}_k(H, H_{\mathrm{ref}})$ is nearly perfectly anti-correlated with f_{spe} ($\rho_{\mathrm{Sp}} = -0.959$, Figure 1). Within each task, f_{spe} predicts LP accuracy with mean Spearman $\rho = +0.420$. Subtracting subspace alignment is therefore an empirically motivated, label-free penalty for candidates whose class-mean variance is absorbed by the generic subspace; this observation is mechanistic evidence rather than a downstream guarantee. We exclude `resnet152` uniformly because reference and target features are available for only one of the eight VTAB datasets; all 168 pairs on the resulting 24-architecture support are computed successfully (see `results/camera_ready/vtab_fspe_24arch_uncentered.csv`).

Alignment and algorithm. For candidate m, let the uncentred feature matrices be $H_{m,T} \in \mathbb{R}^{n_T \times d_m}$ and $H_{m,R} \in \mathbb{R}^{n_R \times d_m}$, both produced by encoder m. Let $V_T, V_R \in \mathbb{R}^{d_m \times k_m}$ contain their leading right-singular vectors, where $k_m = \min\{k, n_T, n_R, d_m\}$ and $k = 20$ by default. With principal angles $\theta_1, \ldots, \theta_{k_m}$,

$$\mathrm{SA}_k(H_{m,T}, H_{m,R}) = \frac{1}{k_m} \|V_T^{\top} V_R\|_F^2 = \frac{1}{k_m} \sum_{i=1}^{k_m} \cos^2 \theta_i.$$

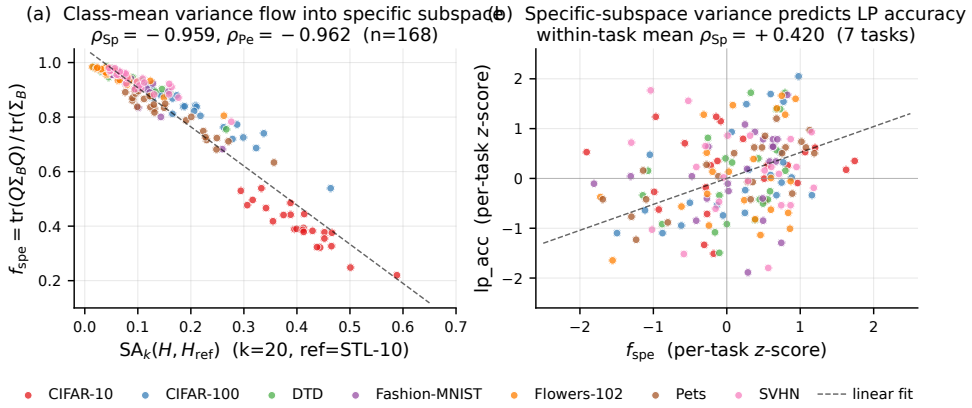


Figure 1: (a) Subspace alignment SA_k between a candidate encoder and the reference (STL-10) is nearly perfectly anti-correlated with the fraction of class-mean variance in the architecture-specific subspace, f_{spe}. (b) Within-task, f_{spe} predicts linear-probe accuracy.

The complete computation is: extract target and reference features with the same candidate; compute their uncentred SVDs; retain the two top-k_m bases; compute SA_k from their principal angles; compute RankMe r_{eff} from the target singular values; clip r_{eff} and SA_k below at 10^{-12}; and return the scalar score below. Candidate feature dimensions may differ, because no cross-candidate subspaces are compared.

Method family. Subspace-Corrected Ranking is the family

$$\mathrm{SCR}(H, T; \beta, g) := \frac{\log r_{\mathrm{eff}}(H) + \beta \cdot g\big(\mathrm{SA}_k(H, H_{\mathrm{ref}})\big)}{\log C(T)},$$

where g is a monotonically decreasing function and $\beta \geq 0$. Choosing $g(s) = -\log s$ with $\beta = 1$ gives

$$\mathrm{SSR}^{\ddagger}(H, T) := \frac{\log r_{\mathrm{eff}}(H) - \log \mathrm{SA}_k(H, H_{\mathrm{ref}})}{\log C(T)},$$

and $g(s) = 1 - s$ gives an additive variant. For per-task ranking, the common positive factor $1/\log C(T)$ may be omitted without changing the selected model. Both forms use target–reference relational geometry and are jointly basis-invariant as in Proposition 2. We adopt $\mathrm{SSR}^{\ddagger}$ as the default because $g(s) = -\log s$ and $\beta = 1$ introduce no tuned correction weight.

Choice of reference dataset. All main experiments use STL-10 as a practical, not theoretically optimal, reference. It is small, contains diverse natural objects, is supported by every candidate, and is disjoint from the seven evaluated VTAB targets. We recommend one generic, target-disjoint reference by default. On the complete 24-architecture support, STL-10 gives VTAB regret 0.061, while a target-disjoint leave-one-target-out median over four predetermined references gives 0.055. The submitted Flowers-102 self-reference result is withdrawn. Because the modest ensemble difference is based on seven tasks and incurs approximately linear reference-side cost, we retain a single reference as the default.

Cross-pool evaluation. We use common support across the zero-label methods: MV (5 tasks), VTAB (7 non-reference tasks), and EX (10 tasks), for 22 pool–task units in total. STL-10 is excluded from every method because it is the reference for $\mathrm{SSR}^{\ddagger}$. On this support, $\mathrm{SSR}^{\ddagger}$ has mean regret 0.069, versus 0.173 for RankMe and 0.137 for uniform random selection. The paired $\mathrm{SSR}^{\ddagger}$-minus-RankMe difference is -0.1043 (unit bootstrap 95% CI $[-0.169, -0.046]$, $p = 0.00016$). Official classification DiSCO is reported separately because it uses target labels.

Dependence-aware uncertainty. The 22 observations contain 10 unique target names, eight repeated across pools. Resampling all rows sharing a target name as one cluster gives 95% CI $[-0.167, -0.040]$ and two-sided $p = 0.00118$ (100,000 resamples); paired Wilcoxon gives $p = 0.00254$. Thus the conclusion remains supported after grouping repeated target names.

Table 7: Zero-label cross-pool comparison on 22 common-support pool–task units. DiSCO is excluded because official classification DiSCO uses target labels.

Method	Mean regret	Label budget	Role
Random	$+0.137$	zero	reference
Flat-RankMe	$+0.173$	zero	target spectrum
Flat-$\mathrm{SSR}^{\ddagger}$	$\mathbf{+0.069}$	zero	target–reference geometry
Paired $\mathrm{SSR}^{\ddagger}$ vs. RankMe: $\Delta = -0.1043$; target-clustered 95% CI $[-0.167, -0.040]$, $p = 0.00118$.			

Scope. The Marchenko–Pastur lemma characterises RankMe under the stated isotropic model; we do not claim a theoretical bound on linear-probe accuracy in terms of $\mathrm{SSR}^{\ddagger}$. The geometric premise is empirical, reference selection is heuristic, and performance relative to label-dependent methods is pool-dependent. Our claim is specifically that $\mathrm{SSR}^{\ddagger}$ corrects RankMe's cross-family width confound on the evaluated heterogeneous common support.

Negative results. Two failure modes delimit the scope of $\mathrm{SSR}^{\ddagger}$. First, including TwoNN intrinsic dimensionality [6] as $\log r_{\mathrm{eff}}/(\mathrm{UID} \cdot \log C)$ does not improve cross-architecture ranking: the pool-level mean $\Delta\rho(\mathrm{SCR}_{\mathrm{uid}} - \mathrm{SSR}^{\ddagger}) = -0.538$ (95% CI $[-0.782, -0.270]$) on MV pool's multi-variant subset, and each architecture family individually shows negative $\Delta\rho$. Intrinsic-dimension proxies do not substitute for subspace alignment. Second, on ViT-only sub-pools where candidates differ only in pretraining recipe of the same backbone family, SA_k profiles are similar but uncorrelated with linear-probe accuracy; $\mathrm{SSR}^{\ddagger}$ underperforms RankMe by $\Delta\rho = -0.26$ (95% CI $[-0.43, -0.07]$, $p = 0.010$). This is a within-family limitation that the second stage of the two-stage pipeline (RankMe within selected architecture, §6) addresses by construction.

6 Two-Stage and Mixed-Budget Pipelines

Algorithm 1 Two-stage SF-MTE pipeline

1: **Input:** candidate pool $\{(a, V_a)\}$; target features H_T; reference H_{ref}.
2: **Stage 1 (architecture).** For each architecture a, compute stage-1 score s_1 on a canonical representative $\bar{v}_a \in V_a$. Set $a^* = \arg\max_a s_1(\bar{v}_a, T)$.
3: **Stage 2 (variant).** For $v \in V_{a^*}$, compute stage-2 score s_2. Return $v^* = \arg\max_v s_2(v, T)$.

Cost and quality. On the five MV tasks, flat scoring evaluates all ~ 119 variants, whereas the two-stage pipeline evaluates $|\{a\}| + \overline{|V_{a^*}|} = 35 + 3 = 38$ candidates on average (range 37–39). Table 8 compares strategies. The label-free pipeline (S5: $\mathrm{SSR}^{\ddagger} \rightarrow$RankMe) has regret $+0.102$, close to flat $\mathrm{SSR}^{\ddagger}$ ($+0.098$), at approximately one third of its forward-pass evaluations. We treat this as a proof of concept on five tasks, not as a general performance or cost guarantee.

Table 8: Pipelines on MV pool multi-variant (35 archs, 119 variants, 5 tasks). Mean LP, regret, 95% task-level bootstrap CI on regret, and forward-pass cost (number of (variant, task) feature evaluations per task). The evaluated label-free two-stage pipeline costs $35 + 3 = 38$ forward evaluations on average; the mixed K=35 sweep uses the recorded $K + 4 = 39$ accounting.

Strategy	Mean LP	Regret	95% CI	Cost
S1 Random	0.627	$+0.194$	$[+0.118, +0.289]$	1.0
S2 Flat-RankMe	0.502	$+0.319$	$[+0.149, +0.509]$	118.8
S3 Flat-$\mathrm{SSR}^{\ddagger}$	0.723	$+0.098$	$[+0.048, +0.148]$	118.8
S4 2S RankMe$\rightarrow$RankMe	0.677	$+0.144$	$[+0.093, +0.220]$	38.0
S5 2S $\mathrm{SSR}^{\ddagger}$ **$\rightarrow$RankMe (proposed, label-free)**	$\mathbf{0.719}$	$\mathbf{+0.102}$	$[\mathbf{+0.044}, \mathbf{+0.153}]$	**38.0**
2S H-score$\rightarrow$RankMe (mixed, K=35 labels at stage 1)	$\mathbf{0.803}$	$\mathbf{+0.018}$	—	**39.0**

Mixed-budget variant. As an exploratory extension on the same five MV tasks, we replace $\mathrm{SSR}^{\ddagger}$ at stage 1 with H-score [2] on K labelled architecture representatives and retain RankMe at stage 2.

The observed regrets are 0.089 at $K{=}5$ and 0.018 at $K{=}35$ (Figure 2). These points illustrate a possible label–compute trade-off on this benchmark; the small task set does not support a general deployment recommendation.

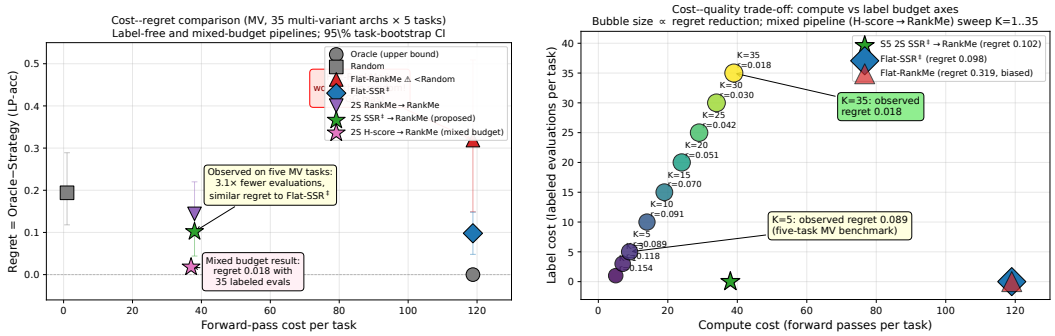


Figure 2: Five-task MV proof of concept. (left) Observed regret versus forward evaluations for the evaluated flat and two-stage strategies. (right) Observed mixed-pipeline regret versus stage-1 label budget K.

7 Discussion

Applicability. On the complete VTAB support (168 architecture–task pairs), SA_k negatively correlates with the fraction of class-mean variance outside the reference subspace ($\rho_{\mathrm{Sp}} = -0.959$). Within a family, however, subspace variation across pretraining recipes does not reliably track linear-probe accuracy. We therefore use $\mathrm{SSR}^{\ddagger}$ for cross-family selection, RankMe for within-family selection, and $\mathrm{SSR}^{\ddagger}$ $\rightarrow$RankMe for mixed multi-variant pools.

Pool dependence and DiSCO. Relative performance varies by pool, particularly on SSL and EX. Official DiSCO uses target ground-truth labels for class-conditional covariances, whereas $\mathrm{SSR}^{\ddagger}$ uses no target labels. We report DiSCO as a label-dependent comparator without claiming uniform superiority.

Limitations. Reference selection is heuristic, and subspace alignment has no formal downstream-accuracy guarantee. On seven VTAB targets, STL-10 and the target-disjoint ensemble yield regrets 0.061 and 0.055; this limited evidence and the added reference cost do not establish a better default. The two-stage cost reduction is a five-task MV proof of concept, not a general guarantee.

8 Conclusion

RankMe is seriously confounded by feature width in heterogeneous architecture pools: under the stated isotropic model its effective rank scales as $\Theta(d)$ and can rank width rather than transfer quality. $\mathrm{SSR}^{\ddagger}$ addresses this specific failure by augmenting the target spectrum with the target–reference subspace relation computed through the same candidate encoder. On the complete common support of 22 pool–task units, it reduces mean regret from 0.173 to 0.069 (difference -0.1043; target-clustered 95% CI $[-0.167, -0.040]$, $p = 0.00118$). The result supports $\mathrm{SSR}^{\ddagger}$ for cross-family zero-label selection and RankMe for within-family selection, while leaving reference design, downstream guarantees, pool-dependent behavior, and broader two-stage validation as open problems.

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A Marchenko–Pastur and Isotropy

Proof of Lemma 1. Write $r_{\text{eff}}(H) = \exp\big(H_{\text{ent}}(\mathbf{p})\big)$, where RankMe uses $p_i = \sigma_i / \sum_j \sigma_j$. Let $\lambda_i = \sigma_i^2/(n\sigma^2)$ and let the empirical measure include all d eigenvalues, including the zero eigenvalues when $d > n$. By the Marchenko–Pastur law [18], this measure converges almost surely to μ_γ. Define $S_d = d^{-1}\sum_j \sqrt{\lambda_j}$ and $m_{1/2} = \int \sqrt{x}\,\mathrm{d}\mu_\gamma(x)$. The common scale $\sigma\sqrt{n}$ cancels from RankMe, and the exact finite-d identity is

$$H_{\text{ent}}(\mathbf{p}) = \log d + \log S_d - \frac{1}{2S_d}\frac{1}{d}\sum_i \sqrt{\lambda_i}\log\lambda_i.$$

Consequently, with $0\log 0 = 0$, the Marchenko–Pastur limit gives

$$H_{\text{ent}}(\mathbf{p}) = \log d + \log m_{1/2} - \frac{1}{2m_{1/2}}\int \sqrt{x}\log x\,\mathrm{d}\mu_\gamma(x) + o(1).$$

The required moments are finite for every fixed positive aspect ratio, including regimes with support reaching zero or an atom at zero. Hence $r_{\text{eff}}(H)/d \to c(\gamma) > 0$ and $r_{\text{eff}}(H) = \Theta(d)$. □

Empirical near-isotropy diagnostic. On the VTAB architecture–task pairs for which the diagnostic is available, the class-covariance statistic $\eta_\Sigma = \mathrm{Var}_c[\mathrm{tr}(\Sigma_c)]/(\mathbb{E}_c[\mathrm{tr}(\Sigma_c)])^2$ is small for most pairs (median 0.022, mean 0.035, and 93.3% below 0.1). This is descriptive evidence that the isotropic approximation is relevant to these representations; it is not a bound on RankMe's ordering or a theorem for other rotation-invariant scores.

Empirical width association. Figure 3 shows the per-architecture mean r_{eff} plotted against d on log-log axes for MV, VTAB, and EX pools. In MV and EX (both using full-dataset features), the Spearman rank correlation between r_{eff} and d is $\rho_{\text{Sp}} = +0.70$ and $+0.86$ respectively. VTAB uses n=1000 subsampled features; for large-d architectures ($d > n$), r_{eff} is capped by n, producing a lower band ($\rho_{\text{Sp}} = +0.41$). These finite-pool observations are consistent with, but do not by themselves prove, the asymptotic lemma.

B MV Pool (Multi-Variant Pool)

Setup. The MV inventory contains 132 ImageNet-1k pretrained variants; the reported architecture-selection and two-stage comparisons use the fixed 119-variant support belonging to 35 families with at least two variants. All are sourced from timm [30] and torchvision and evaluated on 5 target tasks (CIFAR-10, CIFAR-100, DTD, EuroSAT, SVHN). Architecture families span four broad classes: (i) VGG [25] (3 variants, d=4096), ResNets [10] (8 variants, $d \in \{512, 2048\}$), DenseNets (4 variants, $d \in \{1024, 2208\}$); (ii) EfficientNet [26] (4 variants, $d \in \{1280, 1408, 1536\}$), MobileNet/RegNet ($d \in \{960, 1512\}$); (iii) ViT [5]/DeiT (6 variants each, $d \in \{192, 384, 768\}$); (iv) ConvNeXt [16]/Swin (4 variants, d=768) and others. Penultimate dimensions range from d=192 (ViT-tiny, DeiT-tiny) to d=4096 (VGG-16), a $21\times$ spread. LP ground truth uses ℓ_2-regularised logistic regression (C=1) on an $80/20$ split, identical to the VTAB protocol.

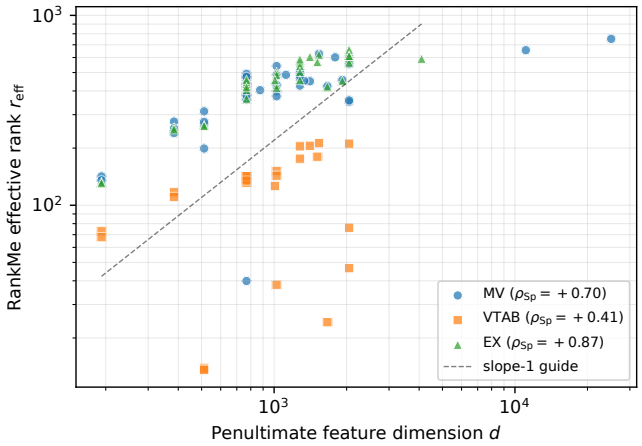


Figure 3: Effective rank $r_{\text{eff}}(H)$ vs. penultimate dimension d on log-log axes, averaged across tasks per architecture. All three pools show positive rank association between r_{eff} and d, consistent with Lemma 1. Dashed/dotted slope-1 lines are visual references, not fitted asymptotic laws. VTAB uses n=1000 subsampled features, so $r_{\text{eff}} \leq n$ creates the lower band when $d > n$.

Table 9: Architecture families in the MV pool with variant counts and penultimate dimensions. The $21\times$ spread in d is the primary driver of the width confound.

Family	Backbone class	# variants	d
VGG	CNN	3	4096
ResNet	CNN	8	512, 2048
DenseNet	CNN	4	1024, 2208
EfficientNet	CNN	4	1280–1536
MobileNetV3	CNN	2	960
RegNet	CNN	3	368–1512
ViT	Transformer	6	192, 384, 768
DeiT	Transformer	6	192, 384, 768
ConvNeXt	ConvNet	4	768
Swin Transformer	Transformer	4	768
Other (MnasNet, etc.)	CNN / hybrid	5	1280–1984

Architecture families and variant counts.

Width confound detail. Per-task $\rho_{\text{Sp}}(\text{RankMe}, d)$ ranges from $+0.50$ to $+0.72$ (mean $+0.61$; all $5/5$ tasks positive), confirming strong width–score correlation even within this single pool. RankMe selects `vgg16` (d=4096) on all 5 target tasks; the oracle is `convnext_base` or `densenet201` depending on the task. Mean RankMe regret is $+0.319$, which is $+0.125$ worse than uniform random selection ($+0.194$). Removing the 3 VGG variants (about 2.3% of the pool) drops Flat-RankMe regret to $+0.144$, illustrating how a single high-d outlier family drives the confound. The observed $\text{SSR}^{\ddagger}$ regret is $+0.098$ both with and without the VGG variants; we treat this as an empirical robustness check rather than a general invariance claim.

Multi-variant structure and two-stage evaluation. The pool contains 35 architecture families with a mean of 3.8 variants per family (range 1–8). The 35 families $\times 5$ tasks multi-variant subset (using the 119 variants belonging to families with ≥ 2 members) is the only subset on which the two-stage pipeline can be evaluated: stage 1 selects the best architecture family using one canonical variant per family, and stage 2 searches within the selected family. The observed $3.1\times$ forward-pass reduction relative to flat evaluation (Sec. 6) follows from the recorded mean cost $35 + 3 = 38$ versus 118.8.

CNN-only sub-pool. To confirm the width confound is not a ViT-vs-CNN artefact, we evaluate the 78 CNN-class variants (removing all ViT, DeiT, Swin families). RankMe selects `vgg16` on $4/5$ tasks (on DTD it selects `densenet201`, d=2208, the next-highest d), and $\rho_{\text{Sp}}(\text{RankMe}, d) = +0.57$ on the CNN-only pool. The confound therefore persists within a homogeneous architecture class,

ruling out the hypothesis that it is driven by the qualitative ViT/CNN feature-structure difference rather than width itself.

C SSL/Multimodal Pool

Setup. We extract penultimate-layer features from 8 SSL/multimodal encoders: ResNet-50/101, MoCo-v3-R50, ConvNeXt-B, ViT-B/16, CLIP-ViT-L/14, DINOv2-ViT-B/14, DINO-ViT-S/16. Feature dimensions $d \in \{384, 768, 1024, 2048\}$; no VGG-class architectures. Evaluation uses 29 downstream tasks covering VTAB, MedMNIST variants, and standard transfer benchmarks. LP ground truth uses the same $80/20$ logistic-regression protocol as MV pool.

Width confound. Per-task $\rho_{\mathrm{Sp}}(\mathrm{RankMe}, d)$ averages $+0.815$ (range $[+0.45, +0.95]$; $27/29$ tasks positive). Flat-RankMe selects a d=2048 ResNet-class encoder on $25/29$ tasks while the oracle is CLIP-ViT-L/14 on $11/29$ tasks and ConvNeXt-B on $8/29$. Mean RankMe regret is $+0.103$ vs. random $+0.066$ ($26/29$ tasks suboptimal).

Label-dependent comparison. Replacing the submission's proxy implementation with official PCA-128 DiSCO gives regret 0.045 for G=6 and 0.042 for G=8 on the available 29 SSL tasks. The withdrawn proxy gives 0.026, while official full-feature scoring gives $0.057/0.047$; these are different algorithms or dimensionality settings, not reruns of one protocol. The values $0.077/0.071$ in an early response draft match no retained output and were a transcription error. Because official classification DiSCO uses target ground-truth labels to form class-conditional covariances, we report it separately from the zero-label $\mathrm{SSR}^{\ddagger}$–RankMe comparison and make no general superiority claim.

Table 10: Official PCA-128 classification DiSCO on each pool. G is the number of consecutive, near-equal spectral groups. DiSCO uses target labels and is therefore not included in the zero-label comparison.

Pool	Tasks	G=6 regret	G=8 regret
MV	5	0.146	0.146
VTAB (STL-10 excluded)	7	0.067	0.040
EX	10	0.045	0.049
SSL	29	0.045	0.042

D VTAB Pool

Setup. The final common pool contains the 24 timm-pretrained architectures with complete features and LP values on all 8 VTAB tasks (CIFAR-10/100, DTD, Fashion-MNIST, Flowers-102, Pets, STL-10, SVHN). The original manifest also listed ResNet-152, but only one of its eight feature files was available, so it is excluded uniformly rather than allowing the candidate set to vary by task. STL-10 features serve as the reference for SA_k and are excluded as a target in the common-support analysis.

Reference-task and k ablation. The audited protocol uses uncentred SVD, RankMe's singular-value normalisation, and excludes STL-10 as a target. With k=20, STL-10 as the single reference gives regret 0.061 over the remaining seven tasks on the complete 24-architecture support. The submitted Flowers-102 self-reference result is withdrawn because the reference coincided with a target. Earlier ablations place the stable range near $k \in [10, 50]$; we use the midpoint k=20 without tuning on target labels.

Multi-reference robustness. For a target-disjoint evaluation, we omit any reference whose dataset name equals the current target and take the median alignment across the remaining members of the predetermined set (STL-10, CIFAR-10, Flowers-102, Pets). This leave-one-target-out ensemble gives regret 0.055, versus 0.061 for STL-10 alone. The modest seven-task difference does not offset the ensemble's approximately M-fold reference-side cost, so we retain one generic target-disjoint reference as the default and view reference design as an open problem.

E EX Pool and Stage-1 Comparisons

EX pool (51-variant $\times$ 10-task, supervised). 51 ImageNet-1k pretrained variants from 35 canonical architectures (including VGG-16, d=4096; $d_{\max}/d_{\min}$=21.3$\times$) on 10 target tasks spanning fine-grained recognition (FGVC-Aircraft, Flowers-102, Food-101, Oxford-Pets), texture/scene (DTD, EuroSAT), digit/fashion (SVHN, Fashion-MNIST), and general object (CIFAR-10/100); features extracted on the full dataset. Flat-SSR‡ regret is $+0.060$, compared with $+0.144$ for RankMe. Official, target-label-dependent DiSCO obtains $+0.045$ with G=6 and $+0.049$ with G=8; it is not part of the zero-label comparison.

Table 11: Reproduced MV-pool flat comparison (same data as Table 5, repeated here for reader convenience). RankMe, LogME, and H-score select `vgg16` on every task; SSR‡ is robust.

Method	With VGG (regret)	Without VGG (regret)	Type
Random	$+0.194$	$+0.193$	—
Flat-RankMe	$+0.319$	$+0.144$	label-free
Flat-LogME	$+0.319$	$+0.550$	label-based
Flat-H-score	$+0.319$	$+0.470$	label-based
Flat-NCTI	$+0.162$	$+0.155$	label-based
Flat-SSR‡	$\mathbf{+0.098}$	$\mathbf{+0.098}$	**label-free**
Flat-LEEP	$+0.038$	$+0.038$	label-based

Spectral truncation. Recomputing r_{eff} on the top-D singular values is a useful diagnostic: at D=256 on MV, the width correlation decreases and flat regret approaches that of SSR‡. We do not promote truncation as a stage-1 alternative because candidates with $d < D$ remain untruncated and the cutoff introduces an additional design choice.

F Ablations and Negative Results

Withdrawn exploratory rules. We do not retain the earlier $d_{\max}/d_{\min}$ threshold for choosing between SSR‡ and DiSCO: the latter is label-dependent, and the apparent crossover was computed with the superseded proxy implementation. We likewise omit cross-pool tuning claims for the additive SCR variant because they are unnecessary for the parameter-free score evaluated in the main result.

Algebraic form of the correction. Within the $-\alpha \log \text{SA}_k$ family on MV pool, $\alpha \in [0.5, 2]$ all yield similar regret ($+0.093$ to $+0.162$); the inverted-direction control $-\log(1 - \text{SA}_k)$ produces regret $+0.366$, confirming that the subtraction direction matters. We adopt $\alpha = 1$ for parsimony.

Per-task breakdown of Flat-RankMe on MV pool. Flat-RankMe selects `vgg16` on every task; the VGG-16 d=4096 penultimate layer dominates r_{eff} regardless of target. Flat-SSR‡ selects task-appropriate architectures (ConvNeXt, ResNet, EfficientNet variants).

Task	Oracle	Random	Flat-RM	Flat-SSR‡	RM picks	SSR‡ picks	Oracle arch
CIFAR-10	0.970	0.785	0.550	0.870	vgg16	resnet50	convnext_base
CIFAR-100	0.825	0.484	0.180	0.670	vgg16	resnet152	convnext_base
DTD	0.740	0.564	0.445	0.720	vgg16	convnext_base	convnext_base
EuroSAT	0.955	0.881	0.765	0.905	vgg16	efficientnet_b3	densenet201
SVHN	0.615	0.430	0.570	0.450	vgg16	efficientnet_b3	densenet169

Negative result 1: UID hurts within-family variant ranking. Including TwoNN intrinsic dimensionality [6] in the denominator ($\text{SCR}_{\text{uid}} = \log r_{\text{eff}}/(\text{UID} \cdot \log C)$) does not improve variant ranking. On MV pool's multi-variant subset, the pool-level mean $\Delta\rho(\text{SCR}_{\text{uid}} - \text{SSR}^{\ddagger}) = -0.538$ (95% CI $[-0.782, -0.270]$); each architecture family individually shows negative $\Delta\rho$.

Negative result 2: SSR‡ degenerates on ViT-only sub-pools. When candidates are restricted to ViT-family architectures (ViT, DeiT, Swin variants on a 33-model $\times$ 10-task benchmark), SA_k profiles are similar across pretraining recipes but uncorrelated with LP accuracy; SSR‡ underperforms RankMe by $\Delta\rho = -0.26$ (95% CI $[-0.43, -0.07]$, $p = 0.010$, task-paired bootstrap). In mixed pools containing both ViT and CNN architectures (the deployment scenario), SSR‡ correctly identifies the better architecture.

Within-family ranking and the $\log C$ scaler. Within any single architecture family, $\mathrm{SSR}^{\ddagger}$ provides no additional ranking signal over RankMe ($\Delta\rho \in [-0.48, -0.07]$ across the 6 families on MV pool). The $\log C$ task scaler is convenient but not principled: an oracle scaler $1/(1-\overline{\mathrm{LP}})$ achieves Spearman $\rho = +0.82$ vs. $\mathrm{SSR}^{\ddagger}$'s $+0.49$, and a random per-task scaler achieves $+0.39$. We retain $\log C$ for cross-task aggregation while noting that other per-task scalers would work comparably.

G Experimental Environment and Compute Accounting

Accelerated feature extraction. The MV and EX feature-extraction reruns supporting the corrected DiSCO evaluation used Huawei Ascend NPUs through PyTorch's `torch_npu` extension and the Ascend toolkit. The retained launch script assigns independent candidate-model shards to four NPU devices (indices 0–3), with one extraction process per device and a configured batch size of 64. Models are evaluated in inference mode and features are cached as float32 arrays. This describes the documented rerun configuration; it does not imply that every historical feature cache was generated on the same hardware. The repository also contains a CUDA reproduction guide recommending at least 16 GB of accelerator memory.

Cached-feature analysis. The corrected DiSCO scoring, VTAB linear-probe completion, reference-subspace audit, and bootstrap analyses operate on cached features using CPU-based Python routines with NumPy, SciPy, and scikit-learn. The retained MV/EX DiSCO scoring launcher configures 16 worker processes. Caching separates encoder inference from subsequent scoring and allows repeated metric evaluation without repeating feature extraction.

Reporting scope. The cost comparisons in Section 6 count candidate–task forward evaluations; they are not measured wall-clock speedups, energy savings, or comparisons between NPU and GPU hardware. Reference-feature extraction and caching contribute additional setup cost. Exact accelerator model and memory, host CPU and RAM, software-version snapshots, and complete per-run runtimes were not consistently preserved across the historical runs. We therefore report the retained configuration without claiming a complete project-wide compute budget, and retain a "No" answer to the compute-resources checklist item.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Sections 1 and 7 state the RankMe-specific width-confounding claim, the cross-family scope of SSR‡, the corrected 22-unit result, and the principal limitations. The five-task two-stage result is explicitly described as a proof of concept rather than a general guarantee.

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   Question: Does the paper discuss the limitations of the work performed by the authors?

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   Justification: Section 7 discusses heuristic reference selection, pool-dependent performance, the absence of a formal downstream-accuracy guarantee, within-family degeneration, and the limited five-task evidence for the two-stage pipeline.

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